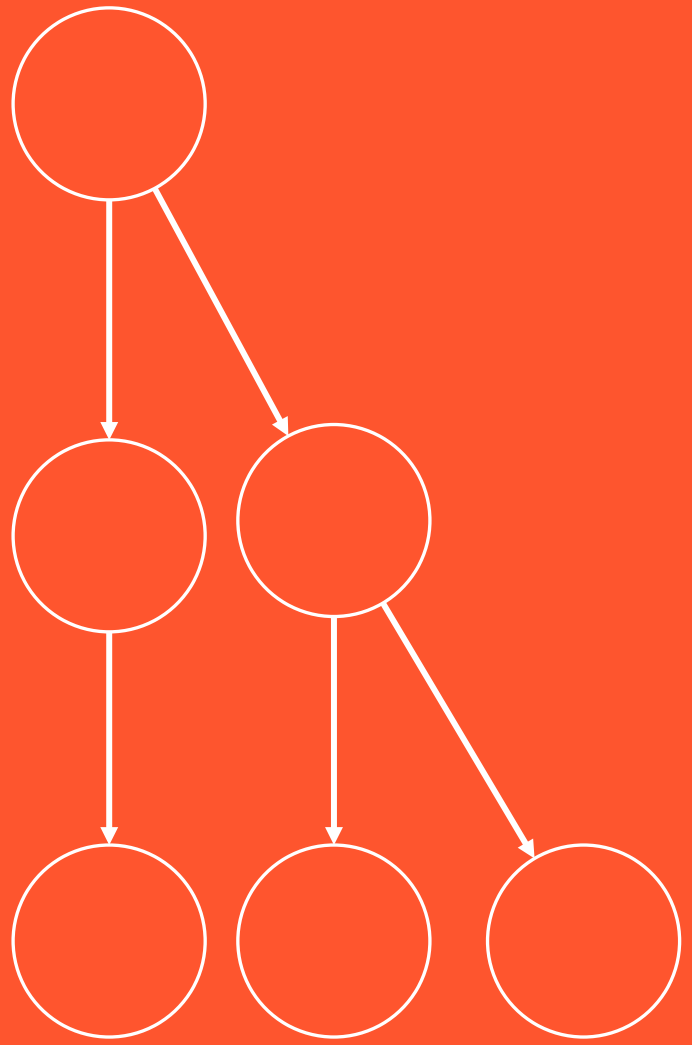


BCOG 100

Social Cognition

Lecture 11.3 — Social Brains

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

Theory of Mind in Machines

In your group, four minutes.

What is theory of mind, and does ChatGPT have it?

How would you know?

New Parts, Not Only More Brain

The **prefrontal cortex** is the cortex at the front of the frontal lobe, ahead of the areas that control movement.

Primate brains: new parts, not only more brain

Larger for body size

monkeys and apes: about twice as large as most mammals' brains, for body size

Denser cortex

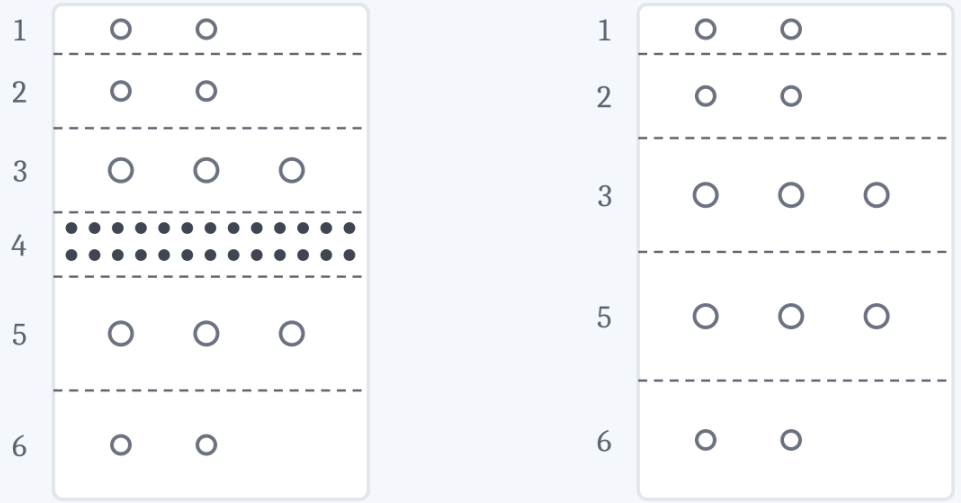
neurons packed more densely than in rodent cortex, with smaller cells

Probably new areas

a set of visual areas; an enlarged region at the back of the parietal lobe, with zones for reaching, grasping, and defense; an area of the premotor cortex

Two kinds of prefrontal cortex

Prefrontal cortex: at the front of the frontal lobe, ahead of the areas that control movement



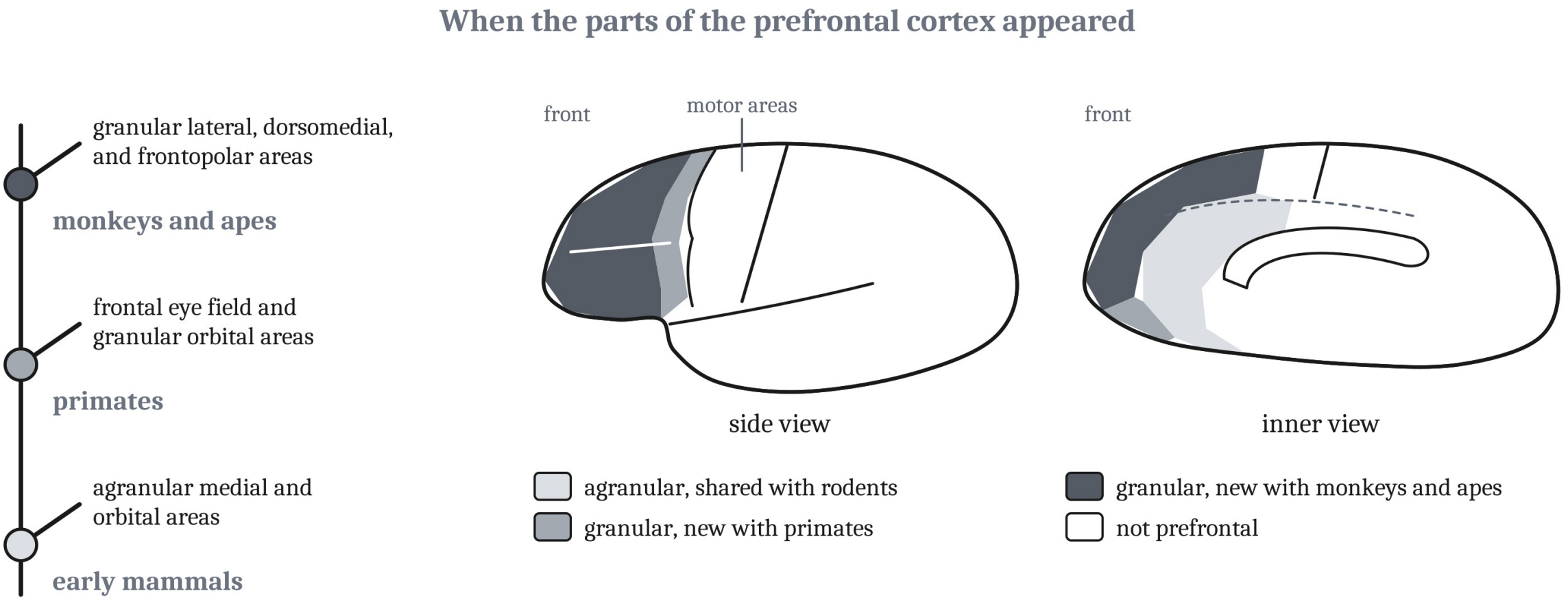
granular
distinct layer 4 of small cells

agranular
no distinct layer 4

schematic; layers numbered from the surface

The Prefrontal Areas, in the Order They Appeared

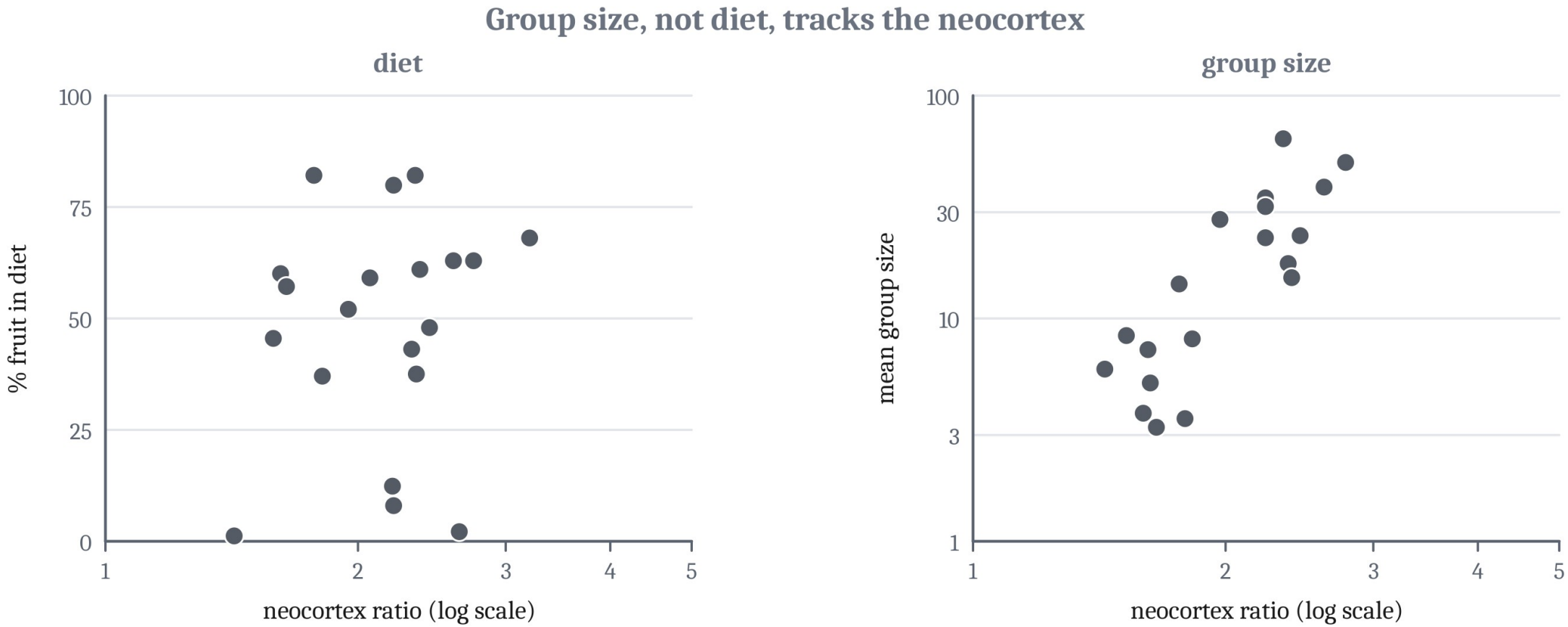
Agranular areas probably go back to early mammals. The first granular areas appeared with primates, or perhaps just before them. Most of the rest appeared with monkeys and apes.



schematic macaque hemispheres; drawn from the account in Preuss and Wise (2022)

The Social Brain Hypothesis

The **social brain hypothesis**: primates living in larger or more complex groups evolved larger brains, because larger brains helped individuals manage life in such groups.



neocortex ratio: the size of the neocortex relative to the rest of the brain

anthropoid genera, read by eye from Dunbar (1998), Fig. 2; approximate

The Same Question, Four Answers

In each study, researchers asked what predicts primate brain size.

Study	Species	Brain measure	Group size	Diet or ranging
Dunbar, 1998	24 anthropoids	Neocortex relative to the rest of the brain	Tracked neocortex size closely	No relation
DeCasien and colleagues, 2017	About 140 primates	Whole brain, body size taken into account	No effect	Fruit-eaters larger, a weak effect
Powell and colleagues, 2017	104 and 144 primates, in two datasets	Whole brain, body size taken into account	Not robust	Home range most consistent; diet in one dataset
Grabowski and colleagues, 2023	128 primates	Whole brain, in a model that lets adaptation lag	Small negative effect	Leaf-eating lowers; mating system raises

What Is Measured, and Change Within a Lifetime

Different brain measures can give different answers. Within one lifetime, the social environment can change the brain.

What is measured

- total brain size
- neocortex relative to the rest
- brain relative to body

These can give different answers. Each stands in for cognition only if cognition varies along one general dimension.

Other animals, other measures

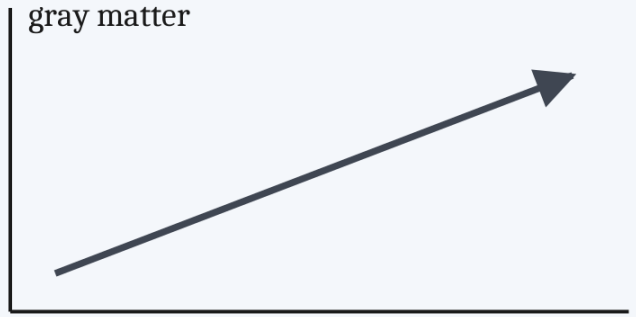
Group size goes with brain size in carnivores, dolphins, and bats in stable groups.

Birds: no clear relation. The largest brains for body size are in species that form long-term pairs or breed cooperatively.

Primates with more forward-facing eyes have larger visual structures and larger brains for their bodies.

Within one lifetime

Macaques housed in groups of one to seven animals



in parts of the temporal and prefrontal cortex; a direction only, no data plotted

The group-size version is not supported by the larger samples; diet and broader social measures have mixed support.

The macaque result is a change within brains, not a test of why brains grew across species.

Faces

In one face patch, recorded in two monkeys, 97% of the cells that responded to visual images responded selectively to faces.

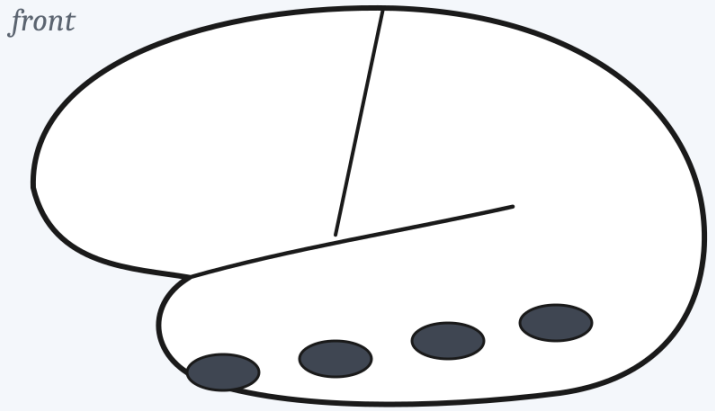
Converging evidence

Upside down, a face is much harder to recognize.

A patch on the underside of the temporal lobe responds far more to faces than to other objects.

Damage there can cause prosopagnosia, an inability to recognize faces.

Face patches in macaques



patches along the temporal lobe respond more to faces than to other objects

schematic; positions approximate

Cells in one patch

share of visually responsive cells that were face selective



one patch, recorded in two monkeys



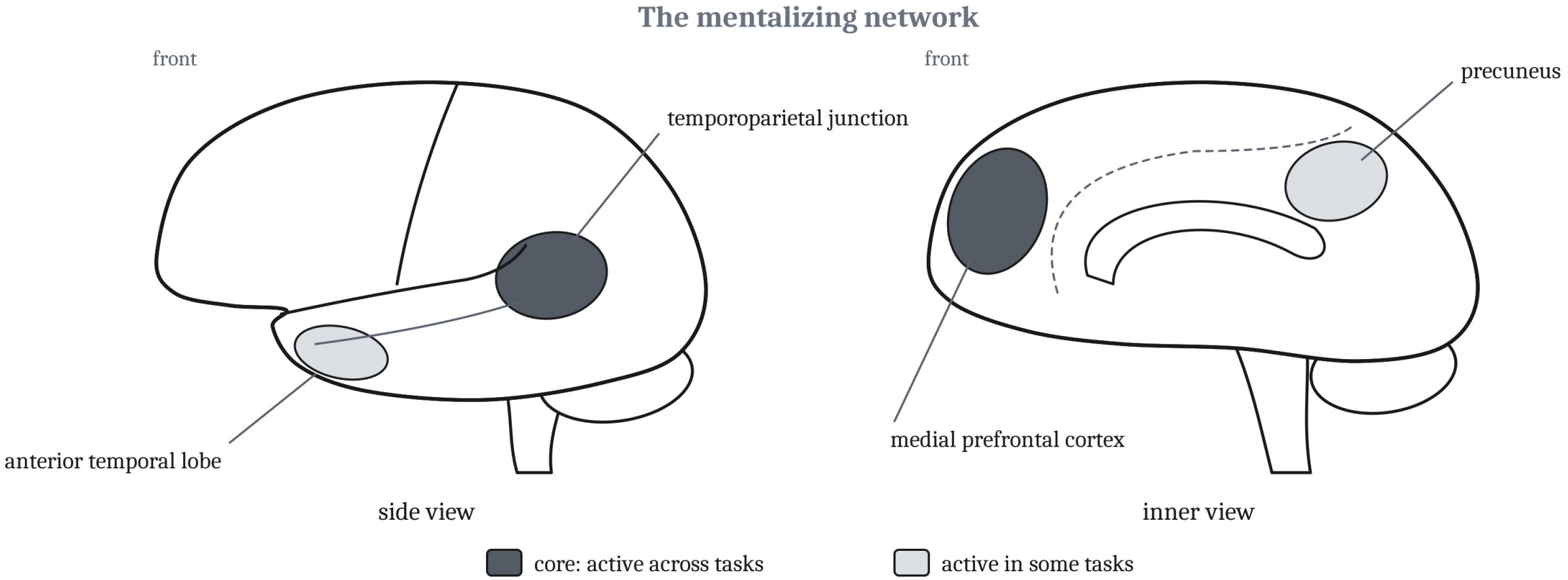
at most 20 to 30%

any region recorded before that study

One patch in two monkeys. Whether the preference was inherited or learned was not tested. Evidence that one region can be highly specialized, not of an inborn face module.

Thinking About What Others Believe

The **temporoparietal junction** is the region where the temporal and parietal lobes meet, on each side of the brain. The **mentalizing network** is the set of brain regions consistently active when people think about others' mental states, at its core the medial prefrontal cortex and the temporoparietal junction on both sides.



schematic; drawn from Schurz and colleagues (2014) and Frith and Frith (2006)

Thinking About Groups

Judging social groups engaged regions also active when people think about others' minds. Whether knowledge about groups is represented like other category knowledge is still open.

Objects

Which is more likely to have six strings: guitars or violins?

Engaged regions of the left frontal and lower temporal lobes, associated with knowledge about objects.

Social groups

Which is more likely to watch romantic comedies: men or women?

Engaged the medial prefrontal cortex and the temporoparietal junction, extending toward the front of the temporal lobes.

The object regions did not respond above baseline when people judged groups.

The authors' conclusion, and two reasons it is still open

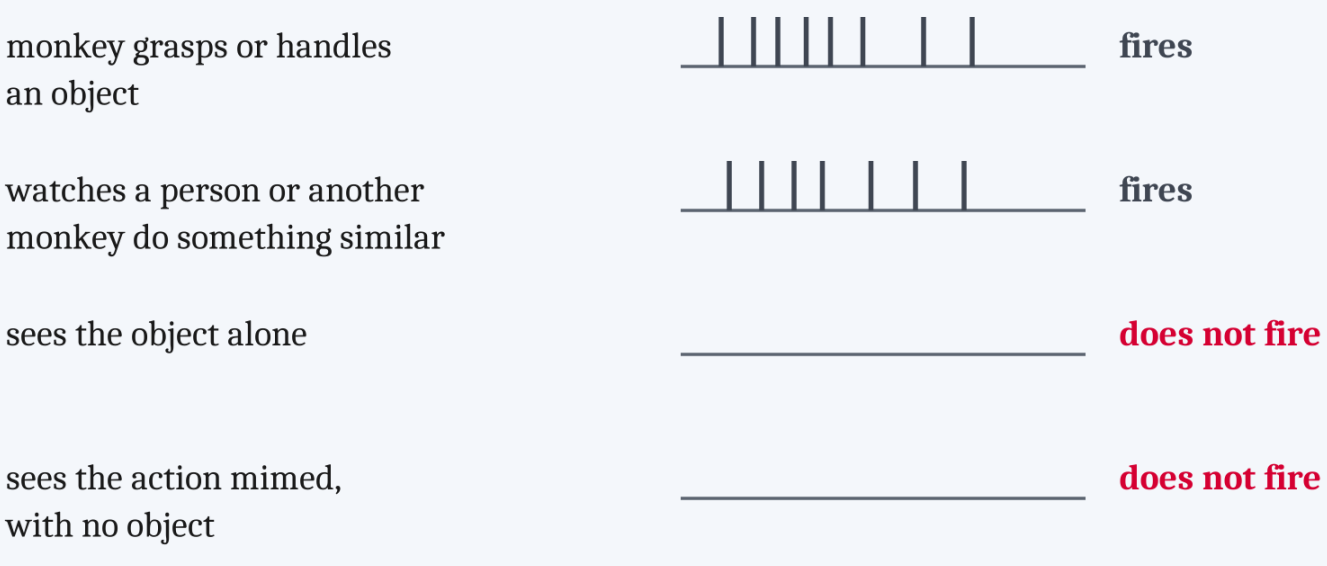
Conclusion: judging social groups has more in common with thinking about minds than with knowing about objects.

But the features about people were mostly preferences, which are mental states themselves.
And the front of the temporal lobes is where, on one account, a semantic hub links what is known about a concept.

Mirror Neurons

A **mirror neuron** is a cell that fires both when an animal performs an action and when it observes a similar action.

One premotor cell in a macaque (schematic)



spike trains drawn for illustration, not recorded data

How common

A minority of the cells recorded: fewer than 10% in the first study, 17% in another. Similar cells were later found in parietal cortex and in other motor areas.

In human beings

In 21 people treated for epilepsy: cells with mirror properties in the supplementary motor area and the medial temporal lobe. The monkey regions were not recorded.

A mirror neuron fires both when an animal performs an action and when it observes a similar action.

Mirror Neurons: Proposed Roles, and Where They Come From

The best-supported roles are modest: telling observed actions apart, and imitation.

Proposed, and what reviewers find

inferring others' goals or intentions
little evidence

autism as a failure of mirror neurons
no compelling evidence

empathy
speculative, largely untested

the evolution of language
speculative, largely untested

Best supported, and modest

telling observed actions apart

imitation: copying the form of another's actions

Autistic people sometimes show stronger mirror responses, and imitate as well as others.

Where the mirror property comes from: the associative account (debated)

doing an action

seeing it

→ Seeing an action while doing it links the two, so a cell that fires for doing comes to fire for seeing.

About 90 minutes of training, making one finger movement while watching the opposite one, reversed the usual mirror response.

Many researchers accept the account. The discoverers argue for a broader mirror mechanism.

What It Takes to Live Among Others

What is known about social brains is mostly description rather than explanation.

The problems

Slow lives and long dependence: someone must care for the young.

Kin selection: help to relatives.

Repeated encounters: help between unrelated animals (evidence thinner than often claimed).

Game theory: described without minds.

The procedures

Come in steps: tit-for-tat; tracking what others try to do; acting on what a competitor has seen.

False belief passed by about age 4; infants and apes unsettled; five accounts still compete.

Learning from others; stereotypes.

The brains

Mostly description, not explanation.

Brains became larger. That larger groups drove it is not supported by larger samples; diet and other measures mixed.

Some regions are consistently active when people think about others.

Mirror neurons exist; modest roles.

And machines

On standard written tests, they give the answers that someone representing others' mental states would give, though not on every altered story. Whether they reach those answers the way people do is still open.

To Discuss: A Headline About the Social Brain

In your group, five minutes.

A headline reads: “Scientists find the brain cells that let us understand each other”.

The story says that one region becomes more active when people think about others, and that some cells fire both when an animal acts and when it watches another act.

What would we want to know before believing the headline? Write a more accurate one.

Image Sources (1 of 2)

Slide	Image	File	License	Source
3	New parts of primate brains, and two kinds of prefrontal cortex	fig-granular.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
4	When the parts of the prefrontal cortex appeared	fig-prefrontal-layers.png	CC BY 4.0	Jon Willits — from the Module 11 reading, drawn from the account in Preuss and Wise (2022) · own work
5	Group size, not diet, tracks the neocortex, in the first test	fig-social-brain.png	CC BY 4.0	Jon Willits — from the Module 11 reading; redrawn from Dunbar (1998), Fig. 2 · own work
7	What is measured, other animals, and change within a lifetime	fig-measures.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
8	Faces, from converging evidence to single cells	fig-face-patches.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
9	The mentalizing network	fig-mentalizing-network.png	CC BY 4.0	Jon Willits — from the Module 11 reading, drawn from Schurz and colleagues (2014) and Frith and Frith (2006) · own work
10	Judging social groups, and judging objects	fig-groups-objects.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
11	What a mirror neuron does, and where such cells have been found	fig-mirror.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work

Image Sources (2 of 2)

Slide	Image	File	License	Source
12	Proposed roles, supported roles, and the associative account	fig-mirror-roles.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
13	What a morning in a baboon group requires, at three levels	fig-close.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work